

Challenge (Habanero)

- Q1.** A hiker's elevation h in meters above sea level after t hours is given by $h(t) = -2t^2 + 12t + 10$. What is the vertex of this parabola?
- (A) (3, 28)
(B) (3, 16)
(C) (6, 10)
(D) (2, 22)
(E) (4, 18)
- Q2.** A park ranger tracks the population P of deer after x years using the equation $P(x) = 50x^2 - 200x + 400$. What is the axis of symmetry?
- (A) $x = -2$
(B) $x = 0$
(C) $x = 2$
(D) $x = 4$
(E) $x = 6$
- Q3.** The cost C in dollars to transport x tons of cargo is given by $C(x) = 2x^2 + 5x + 100$. What is the minimum cost?
- (A) 100
(B) 105
(C) 110
(D) 120
(E) 125
- Q4.** The height h in meters of a projectile after t seconds is $h(t) = -5t^2 + 20t + 1$. What is the maximum height?
- (A) 21 m
(B) 26 m
(C) 31 m
(D) 36 m
(E) 41 m
- Q5.** A fuel-efficient car's gas mileage M in miles per gallon after x years is modeled by $M(x) = -3x^2 + 15x + 20$. What is the vertex of this parabola?
- (A) (2.5, 32.5)
(B) (3, 30)
(C) (4, 28)
(D) (2, 29)
(E) (1.5, 31.75)
- Q6.** The population P of a certain animal species after x years is given by $P(x) = 15x^2 - 30x + 50$. What are the x -intercepts?
- (A) $x = -1, x = 5$
(B) $x = 0, x = 2$
(C) $x = 1, x = 3$
(D) $x = 2, x = 4$
(E) $x = 1, x = 1$
- Q7.** The depth D in meters of water in a lake after x months is $D(x) = -4x^2 + 12x + 5$. What is the axis of symmetry?
- (A) $x = 0.5$
(B) $x = 1$
(C) $x = 1.5$
(D) $x = 2$
(E) $x = 2.5$
- Q8.** A mountain's elevation E in meters above sea level after x kilometers is given by $E(x) = x^2 - 4x + 10$. What is the minimum elevation?
- (A) 6 m
(B) 8 m
(C) 10 m
(D) 12 m
(E) 14 m

Open Ended Questions (FRQ)

- Q9.** A boat's speed S in kilometers per hour after t hours is given by $S(t) = -t^2 + 6t + 2$. Find the maximum speed and when it occurs.
- Q10.** The growth G of a tree in inches after x years is modeled by $G(x) = 2x^2 + 5x + 1$. Determine the vertex of this parabola and explain its significance in the context of tree growth.

Chef's Tips

- Remind students to show their work and explain their reasoning for FRQs.
- Encourage students to use graphing calculators or software to visualize parabolas.
- Emphasize the importance of labeling axes and providing units when graphing.